

MODEL STEALING: STOLEN MODEL DETECTION

Mark Stephen Thomas
 Universität des Saarlandes
 Saarbrücken, Germany
 Matriculation No: 7081771

Yogesh Kishor Kulkarni
 Universität des Saarlandes
 Saarbrücken, Germany
 Matriculation No: 7083560

1 INTRODUCTION

Training machine learning models requires substantial computational and monetary investment, making them valuable intellectual property. Model stealing attacks allow adversaries to obtain copies or functional derivatives at a fraction of the original cost. In this task, we take the victim’s role: given white-box access to a target CIFAR-100 ResNet-18 classifier and 360 suspect models, our goal is to assign each suspect a continuous stealing confidence score. Stolen models include direct copies, function-preserving transformations, fine-tuned checkpoints, and distilled models. We are evaluated using $\text{TPR}@5\%\text{FPR}$.

2 APPROACH

Our best pipeline (v2) combines nine complementary signals spanning weight-space, behavioral, and representation analysis, aggregated through z-score normalization and sigmoid calibration.

Weight-space signals. (1) *Layer-weighted cosine similarity*: cosine similarity between corresponding parameters, weighted by layer depth (conv1: 0.02, layer1: 0.05, layer2: 0.08, layer3: 0.15, layer4: 0.30, fc: 0.40), since early layers converge to similar filters regardless of training. (2) *FC layer similarity*: cosine similarity on fc.weight and fc.bias only—the strongest single fingerprint, robust even after fine-tuning. (3) *BatchNorm statistics similarity*: cosine similarity on concatenated running_mean and running_var vectors, which reflect the exact training data distribution.

Behavioral signals. (4) *Logit correlation (clean)*: Pearson correlation of flattened logits on 2048 test samples, capturing soft output similarity. (5) *Top-3 agreement*: fraction of test samples where top-3 predicted classes overlap—more sensitive than argmax-only agreement. (6) *Logit correlation (noise)*: Pearson correlation on 1024 fixed Gaussian noise inputs, exposing decision boundary geometry where independent models diverge. (7) *Noise prediction agreement*: argmax agreement on noise inputs, the core ModelDiff signal (3)—stolen models inherit identical decision boundaries on OOD inputs.

Dataset and representation signals. (8) *Train–test loss gap*: Spearman correlation of per-sample losses on training data minus the same on test data. Stolen models show correlated training losses due to shared memorization, producing a positive gap. (9) *Layer4 CKA*: linear Centered Kernel Alignment on layer4 activations (global-average-pooled to $\mathbb{R}^{512}$). CKA is invariant to orthogonal rotation and scaling, catching fine-tuned suspects whose weights diverge but whose learned representations are inherited.

Score calibration. Each signal is z-normalized across all 360 suspects. The normalized signals are combined using the weighted sum in Table 1, then passed through a sigmoid ($\sigma = 3.0$) to produce scores in $[0, 1]$.

3 RESULTS

Table 2 shows our progression on the public leaderboard (30% of models). Each pipeline processes 360 suspects in ~ 5 hours on a Tesla P100-16GB.

Key observations: exact copies (suspects 71, 124, 138, 145, 358) score ≈ 1.0 across all signals. Distilled models (suspects 4, 40, 64) show FC similarity of 1.0 and perfect logit correlation but lower weight-space cosine (~ 0.5), indicating different parameters preserving functional behavior.

Signal	Weight	Signal	Weight
Weighted cosine	0.12	Noise logit corr.	0.08
FC similarity	0.13	Noise agreement	0.12
BN similarity	0.18	Loss gap	0.10
Logit corr. (clean)	0.10	Layer4 CKA	0.15
Top-3 agreement	0.08		

Table 1: Signal weights for the v2 ensemble. BN similarity and Layer4 CKA are promoted as the strongest signals for direct copies and fine-tuned variants respectively.

Method	TPR@5%FPR
v1: 7-signal, layer-weighted cosine, sigmoid calibration	0.5741
v2: 9-signal, +CKA, +noise agreement, +loss gap, +top-3	0.5926

Table 2: Leaderboard scores (TPR@5%FPR). Higher is better.

The noise agreement signal is strongly bimodal (0 vs 1), cleanly separating extraction-based stolen models. Layer4 CKA captures fine-tuned variants that weight cosine misses. Independent models consistently show negative FC similarity and low noise-logit correlation (< 0.4).

4 CONCLUSION

Our multi-signal detection pipeline achieves TPR@5%FPR of 0.5926, demonstrating that stolen models can be reliably identified through combined weight-space, behavioral, and representation analysis. The practical implications of model stealing are significant: stolen models enable adversaries to bypass training costs, undermining ML-as-a-service providers. They facilitate crafting adversarial examples via white-box access to a stolen copy, circumventing usage restrictions, and extracting proprietary training data through inversion attacks. As foundation models become increasingly expensive to train, the economic incentive for theft grows proportionally, making robust detection essential for protecting intellectual property in deployed ML systems.

Code: https://github.com/MARKST-47/TML-26/tree/main/tml26_task2

REFERENCES

- [1] Jagielski, M., Carlini, N., Berthelot, D., Kurakin, A., & Papernot, N. (2020). High accuracy and high fidelity extraction of neural networks. *USENIX Security Symposium*.
- [2] Orekondy, T., Schiele, B., & Fritz, M. (2019). Knockoff nets: Stealing functionality of black-box models. *CVPR*, 4954–4963.
- [3] Shah, H., Park, S.M., Ilyas, A., & Madry, A. (2023). ModelDiff: A framework for comparing learning algorithms. *ICML*.